

GARCH Model

$$AR(1) \Rightarrow X_t = \phi X_{t-1} + w_t$$

← Const. Var. σ^2

Introducing $r_t =$ return on growth rate

$$r_t = \frac{X_t - X_{t-1}}{X_{t-1}}$$

$$\approx \nabla \log X_t$$

$$\begin{aligned} \log(1+r) &\approx r \Rightarrow \log\left(1 + \frac{X_t - X_{t-1}}{X_{t-1}}\right) \\ &= \log\left(\frac{X_t}{X_{t-1}}\right) = \nabla \log X_t \end{aligned}$$

First, ARCH(1) Model

$r_t \Rightarrow$ return

$\sigma_t^2 =$ time varying variance

$w_t \sim$ iid white noise (Gaussian)

$$\begin{aligned} \bullet \quad r_t &= \sigma_t w_t & \sigma_t^2 &= \alpha_0 + \alpha_1 r_{t-1}^2 \\ & & \uparrow & \quad \uparrow \\ & & \text{const.} & \quad \text{AR(1) coeff.} \end{aligned}$$

$\Rightarrow$ Means, if at $t-1$, return is high, in next step, variance would be high too.

Note: $\alpha_0, \alpha_1 > 0$ as $\sigma_x^2 > 0$

If w_t is Gaussian, then

$$r_t = \sqrt{\alpha_0 + \alpha_1 r_{t-1}^2}$$

$w_t \swarrow$ Assume $N(0,1)$

$$r_t | r_{t-1} \sim N(0, \alpha_0 + \alpha_1 r_{t-1}^2)$$

By sq.

$$r_t^2 = \sigma_t^2 w_t^2 \iff w_t^2 \sim \chi(1)$$
$$- \alpha_0 + \alpha_1 r_{t-1}^2 = \sigma_t^2$$

$\swarrow$ centered $\chi^2(1)$

$$r_t^2 - \alpha_1 r_{t-1}^2 - \alpha_0 = \sigma_t^2 (w_t^2 - 1)$$

$$\Rightarrow r_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2 + \eta_t$$

$\uparrow$

AR(1) with mean and Non-Gaussian Noise

What is the Mean & Variance of x_t^2

• Conditionally, $\mathbb{E}[x_t | x_{t-1}, \dots] = 0$

like Markov Process
(only dependent on x_{t-1})

• $\text{Cov}(x_{t+h}, x_t) = \mathbb{E}[x_{t+h}, x_t]$

$h > 0 \quad = \mathbb{E}[\mathbb{E}[x_{t+h} x_t | x_{t+h-1}, \dots]]$

$= \mathbb{E}[x_t \underbrace{\mathbb{E}[x_{t+h} | x_{t+h-1}, \dots]}_0]$

$= 0$

x_t uncorrelated ~~***~~

• $\text{Var}(x_t) = \mathbb{E}[x_t^2]$

$\mathbb{E}[x_t^2] = \alpha_0 + \alpha_1 \mathbb{E}[x_{t-1}^2] + \cancel{\mathbb{E}[v_t^2]}$
 $\hookrightarrow \chi(0,1)$

$\mu = \frac{\alpha_0}{1-\alpha_1}$

$\alpha_0 \Rightarrow$ lower bound variable
 $\alpha_1 \Rightarrow$ ARCH(1) param.

$0 \leq \alpha_1 < 1$

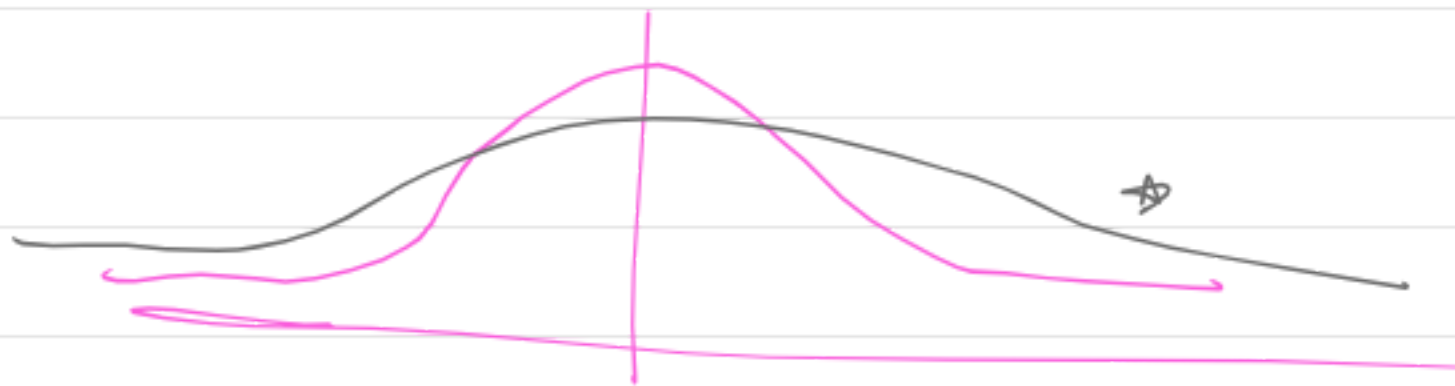
$$\bullet \text{var}(x_t) = E[x_t^2] - (E[x_t])^2$$

$$E[x_t^2] = \frac{3\alpha_0^2}{(1-\alpha_1)^2} \left(\frac{1-\alpha_1^2}{1-3\alpha_1^2} \right)$$

↖ Only works if $0 \leq \alpha_1 < \frac{1}{3}$

Based on the 4th moment of x_t , we see that it is "leptokurtic".

↳ heavier than Gaussian Tails



AR(1) Process

$x_t \leftarrow$ uncorrelated Noise Process with heavier tails than Gaussian when $0 \leq \alpha < 1$

$x_t^2 \leftarrow$ AR(1) process assuming $0 \leq \alpha_1 < \frac{1}{3}$

How to fit this Model?

↳ Conditional MLE
(Maxⁿ Likelihood Est.)

Thus far, the ARCH Model is only for the noise $\rightarrow \sigma_x^2$ and r_x

- ARCH with covariates (linear)

$$X_t = \beta \cdot Z_t + \varepsilon_t$$

$$\beta_1 z_t^{(1)} + \beta_2 z_t^{(2)} \dots$$

Arch Process

$$X = \beta_0 + \beta_1 t + \varepsilon_t$$

Linear Trend

- AR(1) $\times$ ARCH(1)

$$X_t = \phi X_{t-1} + \varepsilon_t$$

- ARCH(p)

σ_x^2 depends on k prev returns

$$\sigma_x^2 = \alpha_0 + \alpha_1 r_{t-1}^2 + \alpha_2 r_{t-2}^2 \dots$$

- GARCH (1,1): $\sigma_x^2 = \sigma^2 \omega_x^2$

$$\sigma_x^2 = \alpha_0 + \underbrace{\alpha_1}_{\Gamma_1} x_{t-1}^2 + \underbrace{\beta_1}_{\Gamma_1} \sigma_{x-1}^2$$

$$\Rightarrow \underset{\uparrow \text{AR}}{h_t^2} = \alpha_0 + (\alpha_1 + \beta_1) \underset{\uparrow \text{MA}}{h_{t-1}^2} + \underset{\uparrow \text{MA}}{v_t} - \beta_1 \underset{\uparrow \text{MA}}{v_{t-1}}$$

$ARMA(p, q)$ $\times$ $GARCH(p, q)$
 Mean Process Heteroscedastic Errors
 X_t ε_t